

CM52066: Research and Development Project Skills

Coursework 2: Project Proposal + Literature, Technology and Data Review

Designing Human–AI Decision Support for Digital Twins of Modernist Heritage Buildings

A Proof-of-Concept Study

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Contents

Declaration of Generative AI Use	2
1 Introduction	3
1.1 Problem Description	3
1.1.1 What This Project Sets Out To Do	3
1.1.2 Scope and Data Ethics	4
1.2 Research Question, Aim and Objectives	5
1.2.1 Objectives	5
1.3 Deliverables	6
2 Literature, Technology and Data Review	7
2.1 Thermal Comfort and the Limits of Threshold-Based Automation	7
2.2 Machine Learning for Personalised Thermal Comfort	8
2.3 Human–AI Decision Support: Interpretability, Trust, Interaction	10
2.4 Heritage Conservation as a Design Constraint	10
2.5 Summary and Implications	11
2.5.1 Limitations of This Review	11
A Project Plan	14
A.1 Methodological Approach	14
A.2 Sprint Plan and Gantt Chart	14
A.3 Tasks and Effort Estimation	16
A.4 Milestones	16
A.5 Risk Register	17
A.6 Resources	17

Declaration of Generative AI Use

In line with University of Bath guidance on Type B coursework, I declare that the following generative AI tools were used in the preparation of this document:

- **Anthropic Claude** was used for: drafting aid to support document structuring, suggest alternative phrasings in early drafts, assisting with L^AT_EX formatting, and helping organise the project plan tables.
- **Grammarly** was used for: spelling and grammar checks throughout.

1 Introduction

1.1 Problem Description

Contemporary building management systems (BMS), including those augmented with digital twin technology, regulate indoor environments by comparing sensor readings against fixed, population-averaged thresholds. For example, If temperature exceeds 25°C, cooling activates. If humidity climbs above 70%, dehumidification engages. The approach is computationally tractable and commercially widespread, and it has served buildings reasonably well for several decades. Cheung et al. (2019) , in a large-scale re-analysis of the ASHRAE Global Thermal Comfort Database II aimed at evaluating Predicted Mean Vote (PMV's) predictive accuracy, report that across more than 100,000 occupant thermal sensation votes (i.e. self-reported perceptions of being too cold, comfortable, or too warm), the Predicted Mean Vote (PMV) model correctly classifies individual thermal sensation in only about one-third of cases. The model retains real utility as a population-level approximation; what it does not do is reliably represent the individual.

The problem intensifies in modernist heritage buildings, where three additional constraints compound the limits of threshold-based automation. Conservation frameworks, notably the Burra Charter (ICOMOS Australia, 2013) and Feilden (2003), require interventions to be minimal, reversible, and explicable. Unreviewed automated actuation satisfies none of these. Heritage sites also serve heterogeneous occupant populations whose comfort preferences span ranges that no single setpoint can accommodate. And the physical fabric of listed buildings restricts both sensor installation and HVAC(Heating, Ventilation, and Air Conditioning) modification, limiting what the system can sense and what it can do. Mazzetto (2024) describes this as a three-way tension between conservation protocols, energy sustainability, and occupant wellbeing.

1.1.1 What This Project Sets Out To Do

This project designs and evaluates a *proof of concept* for a human-in-the-loop decision-support system that supplements, rather than replaces, threshold-based BMS automation. The system reads temperature and humidity data, flags moments where personalised judgement might improve on the threshold rule, and presents an interpretable recommendation to a building occupant or facilities manager who decides what to do. A human-in-the-loop AI cannot resolve the underlying physical disagreement between two occupants who feel differently about the same room; what it can do is help each of them make a better-informed choice about whether to adjust the environment.

The project builds on a prior University of Bath dissertation (Cai, 2025), which collected approximately one year of temperature and humidity data from Connaught Mansions, a listed heritage building in Bath, using Tinytag Ultra 2 data loggers in two flats. Cai characterised the comfort zones and built a Revit BIM model. She did not implement machine learning, recommendation logic, or any mechanism for occupant interaction. That is the layer this project adds, at proof-of-concept scale.

1.1.2 Scope and Data Ethics

The work is scoped as a one-year MSc project running across four sprints (Appendix A). The deliverable is a proof of concept, not a deployment-ready system, and is evaluated through scenario-based testing on existing data rather than live occupant trials.

Data scope and ethics. Both datasets used in this project (the LaSDPC IoT dataset and Cai’s Bath data) are restricted to **temperature and relative humidity readings only**. No occupant identifiers, no personally identifying information, and no fine-grained location data beyond the city of collection are used at any point. Comfort-vote labels, where required, are generated synthetically (§1.1.2) rather than collected from people. This scoping is deliberate: it keeps the project within ethical bounds for an MSc proof of concept and simplifies the ethical-approval pathway, since no human-subjects data is involved.

Dataset characteristics relevant to method choice. The LaSDPC zones show baseline relative humidity in the 57–85% range, leaving temperature manipulation as the primary control lever. Cai’s basement flat at Connaught Mansions sustained >70% RH(Relative Humidity) through autumn 2023 while the second-floor flat performed within accepted comfort bands. Given the population-level limitations of PMV established in Chapter 2, the safer position for both datasets is to treat PMV as a useful prior rather than a trusted predictor and to design the system around the expectation that real comfort will diverge from it.

Use of synthetic data. Whether to generate synthetic comfort labels is a deliberate design decision rather than a default, and it deserves to be argued out rather than assumed. Three options were considered.

Option 1: Collect real labels. The cleanest scientific path would be to recruit occupants and collect comfort votes directly, paired with the environmental readings at the moment of report. This would produce ground-truth data unmatched by any synthetic alternative. It is also incompatible with the proof-of-concept scope and the data-ethics framing in §1.1.2: real-occupant collection requires ethical approval, recruitment effort, and personally identifying information that this project has explicitly chosen not to handle. It is the right option for a future PhD-scale extension, not for an MSc proof of concept.

Option 2: Use existing labelled comfort datasets. Datasets such as the ASHRAE Global Thermal Comfort Database II do contain labelled votes at scale. They are however drawn from buildings whose occupancy patterns, climates, and HVAC controllability differ materially from either the LaSDPC zones or the Bath flats. Training on labels from one distribution and deploying on another reintroduces the very transferability problem Cheung et al. (2019) demonstrate at population level, and one of Boutahri and Tilioua (2024)’s upper-bound assumptions (training and deployment distributions match) would be violated by design.

Option 3: Generate synthetic labels from PMV plus calibrated noise. This is the option chosen, and the choice is justified by what the proof of concept actually has to demonstrate. The technical claim being tested is not “personal comfort can be predicted accurately for a real occupant of Connaught Mansions” (a claim only Option 1 could support); it is “an active-learning loop with SHAP explanations and override logic can be designed and operated coherently within heritage

constraints”. For that claim, synthetic labels are sufficient as long as their noise structure is realistic. Calibrating the Gaussian noise to the variance reported by Cheung et al. (2019) makes the synthetic feedback behave like real occupant feedback in the relevant respect: it disagrees with PMV at roughly the right rate, in roughly the right direction.

The honest limitation is that synthetic labels cannot validate the system’s accuracy for a real occupant; they can only validate that the system’s mechanics work end-to-end. This is acknowledged throughout the dissertation rather than hidden, and a real-occupant trial is named explicitly as future work.

1.2 Research Question, Aim and Objectives

The research question that follows is derived from the literature reviewed in Chapter 2, and specifically from the four gaps identified across §2.1–2.4. The reader is referred to §2.5 for the explicit grounding of the question in those gaps; the question is stated here in the introduction only for orientation.

Research question. How can a digital twin system, using active learning on temperature and humidity sensor data, learn when and how to query occupant comfort preferences in order to provide interpretable, personalised recommendations that support informed human decision-making in a heritage building context?

Aim. To develop and evaluate a proof-of-concept machine learning system that enables meaningful interaction between a building digital twin and its occupants, learning personalised thermal comfort preferences through targeted, minimally intrusive questioning, and presenting recommendations in a form that is transparent and actionable for non-technical decision-makers.

1.2.1 Objectives

- O1. Critical literature review** across four threads: thermal comfort and individual variability, machine learning for personalised comfort, human–AI decision support, and heritage conservation ethics.
- O2. Dataset characterisation.** Map the LaSDPC and Bath datasets restricted to temperature and humidity; establish baseline environmental profiles per zone; assess data quality.
- O3. System architecture design.** Define the active-learning uncertainty threshold, recommendation format, and override / defer / confirm logic of the decision-support interface.
- O4. Proof-of-concept implementation.** Active learning loop, synthetic feedback generator, SHAP-based explanation layer, and a small dashboard that surfaces a recommendation with a plain-language reason.
- O5. Evaluation.** Scenario-based testing on held-out LaSDPC episodes (Appendix A.1). Live human-subjects evaluation is *not* undertaken at MSc proof-of-concept stage; this is discussed as a limitation in §2.5.1.

O6. Critical reflection. Honest accounting of trade-offs between comfort personalisation, energy efficiency, and heritage conservation constraints, with limitations and future directions.

1.3 Deliverables

The project delivers four artefacts:

1. A proof-of-concept software prototype demonstrating the active learning loop and decision-support interface.
2. An evaluation report covering the scenario-based testing results.
3. A dissertation documenting design rationale, implementation, evaluation, and critical reflection.
4. and A public GitHub repository.

2 Literature, Technology and Data Review

This review is organised around four thematic strands:

1. The first establishes the empirical case that threshold-based building automation underperforms at the individual level.
2. The second examines machine-learning responses, leading to the active-learning approach this project adopts.
3. The third considers how such systems should interface with people.
4. The fourth considers heritage conservation as a design constraint.

Within each strand, evidence is presented in order of strength: stronger and larger studies first, smaller studies as complementary evidence. Each strand closes with the gap it exposes; the four gaps together define the contribution this project makes.

2.1 Thermal Comfort and the Limits of Threshold-Based Automation

The framework most building automation relies on is Fanger’s Predicted Mean Vote (PMV) model (Fanger, 1970), codified in ASHRAE Standard 55 (ASHRAE, 2020). PMV has retained practical utility: it has shaped HVAC standards for over five decades and continues to produce sensible aggregate comfort predictions at the population scale, which is why it remains the default in BMS deployments (Arowoia et al., 2024). The relevant question is not whether PMV works in general but how well it works at the level of the *individual*, which is where personalised decision-support has to operate.

The strongest evidence on individual-level accuracy is Cheung et al. (2019), whose re-analysis of the ASHRAE Global Thermal Comfort Database II (>100,000 occupant votes, multiple climates and building types) reports PMV correctly classifies thermal sensation in fewer than 34% of cases. The size of the database and the consistency of the finding across sub-samples make this the strongest available evidence on PMV’s individual-level performance. Two qualifications matter: residential settings are under-represented, and the 34% is a summary statistic rather than a uniform error rate. PMV does not fail randomly; it fails systematically, over-predicting discomfort in naturally ventilated buildings and under-predicting it in mechanically heated ones.

De Dear and Brager (2002), on 21,000 votes from naturally ventilated buildings, found behaviourally adapting occupants tolerated comfort ranges 3–4°C wider than PMV predicts. This is a substantial sample by field-study standards, drawn primarily from Australian and Pacific climates, so its findings transfer with caution to UK heritage settings. Humphreys and Nicol (1998) argue from a smaller corpus that *perceived control* alters reported comfort: two occupants in identical conditions can report different sensations depending on whether they believe they can adjust the environment. The sample is modest and the result has been corroborated rather than independently replicated, so it is treated as complementary evidence. The implication is

bounded but real: a human-in-the-loop AI cannot make two occupants who feel differently feel the same way; what it can do is preserve their perception of agency through a recommendation clearly subject to override.

Karjalainen (2012)’s narrative review reports women on average prefer ambient temperatures 0.5–1.0°C warmer than men, with narrower tolerance in older occupants. As a narrative review it is qualified evidence, but consistent across the studies surveyed.

Donkers et al. (2022, 2024), on the OpenFlat case study with two Dutch homes, observe that occupants of recently certified high-performance buildings still report systematic discomfort. The study’s value lies in its longitudinal feedback record rather than sample size; their findings align directionally with Cheung et al.. Kulkarni et al. (2024) report a similar pattern from an IoT-monitored shared workspace, again on a small sample; their finding that fixed-setpoint systems behave reactively does not depend on sample size and is consistent with Cheung et al.’s larger evidence base. Wolkoff and Kjærgaard (2007) note that low and high relative humidity affect comfort and respiratory health non-linearly: humidity cannot be a simple additive term in any comfort index.

The gap: Threshold-based automation rests on a model that has demonstrated population utility but limited individual-level accuracy. Two responses have emerged: the adaptive comfort model (Nicol and Humphreys, 2002), which broadens tolerance bands but still produces a population-level prediction; and personal comfort models, which abandon group prediction and learn one comfort function per occupant. Neither has been tested under heritage constraints, where the cost of acting on a wrong prediction is conservation damage rather than only discomfort.

2.2 Machine Learning for Personalised Thermal Comfort

Personal comfort modelling, predicting one occupant’s comfort rather than a population’s, has matured within a decade. Abdelrahman et al. (2022) report personal models built from individual feedback outperforming PMV for the individuals they represent. Their occupants were drawn from a Singapore office building with controllable HVAC, so the result establishes feasibility rather than transferability. The relevant question is whether personalisation can work without the labelled training data their study assumed.

A note on terminology. In supervised machine learning, a *label* is the ground-truth answer associated with a data point. For thermal comfort, a label is an occupant’s reported sensation (e.g. “too warm”, “comfortable”) paired with the environmental readings at the moment of report. The literature on personal comfort modelling typically assumes labels are available; the heritage context this project targets does not, since no comfort surveys have been collected.

Boutahri and Tilioua (2024) report 96.7% accuracy using Random Forest on a labelled dataset from one building in Morocco. The methodology is sound, but treating one number from one study as evidence Random Forest is *the* answer would be premature. The accuracy is also an upper bound: it assumes a labelled training set exists, labels are accurate, and training and deployment distributions match. None holds here.

Active learning is one better response to data scarcity. An active learner selects unlabelled

instances whose labels would most improve the model and queries the user only at those moments (Settles, 2009). Tekler et al. (2023) demonstrate it for thermal comfort, reporting accuracy comparable to exhaustive surveys with around 60% fewer interactions on a limited participant set; Tekler et al. (2024) extend it with transfer learning. The published evidence base is narrow: two studies from the same research group in office settings. The methodological argument is strong; empirical evidence for generalisation to heritage settings is what this project tests.

Yang et al. (2024), on a reinforcement-learning controller paired with an LLM human-simulator, are explicit that real-time human feedback is difficult to collect in practice. RL and deep learning are surveyed here despite being rejected because a reviewer in AI will reasonably ask why a more powerful method was not used; an honest comparison answers that. RL needs many interactions to converge, produces opaque policies, and treats occupants as a reward signal, which sits awkwardly with conservation ethics. Deep neural networks need large labelled datasets the heritage context does not provide, and outputs are not interpretable without post-hoc techniques.

Table 2.1: Comparison of machine learning methods for personalised thermal comfort modelling

Method	Reported headline + key limit (with sample where known)	Fit
Reinforcement Learning	Strong on long-run optimisation (Yang et al., 2024); needs thousands of interactions; opaque policies; treats occupants as reward.	Low
Deep Learning / NN	High accuracy on large labelled datasets in the literature; no comfort labels available here; outputs not directly interpretable.	Low
Unsupervised (clustering)	Useful for EDA and zone segmentation; cannot produce action recommendations on its own.	EDA only
Supervised (RF / DT)	96.7% reported in Boutahri and Tilioua (2024) (single building, Morocco); inherently interpretable; needs pre-existing labels.	Medium (base)
Gaussian Process / Bayesian	Principled uncertainty estimates; computationally heavier; less mature tooling for this domain.	Medium
Active Learning	~60% fewer queries vs exhaustive in Tekler et al. (2023) (limited sample, office setting); operates without pre-existing labels; transparent query logic.	High

The gap: Active learning has a plausible methodological fit for personalised comfort modelling without pre-existing labels, but its evidence base sits in commercial buildings with regular occupancy. The key point is one of taxonomy: “commercial” and “heritage” are not mutually exclusive (a commercial gallery can sit inside a listed building, and some heritage buildings have controllable HVAC), but the specific combination of intermittent occupancy, restricted physical fabric, and conservation-mandated review of any actuation has not been the setting for any active-learning comfort study I have located. Whether active learning transfers to that

combination is what this project tests.

2.3 Human–AI Decision Support: Interpretability, Trust, Interaction

A recommendation is only as useful as the human receiving it judges it to be. Amershi et al. (2019) propose eighteen design guidelines for human–AI systems. Their emphasis on scoping services clearly, making it easy to dismiss recommendations, and supporting efficient correction is directly relevant: a conservation officer who cannot meaningfully override the system is not in the loop. The guidelines are general principles rather than a domain specification, so this project treats them as constraints to test against, with the audit-trail requirement of heritage settings layered on top.

Lundberg and Lee (2017) introduced SHAP, a feature-attribution method that has become close to default for tree-based models. Two limits matter here. When inputs are correlated, as temperature and humidity are, SHAP can attribute importance to features that are merely proxies for others. And a numerically correct attribution is not an intuitive explanation: “humidity contributed +0.31 to the uncomfortable prediction” is meaningful to a data scientist but opaque to a building manager. This project treats SHAP as a source for plain-language generation: a sentence such as “humidity has been above 75% for 40 minutes and the last two votes were negative” is what the user sees; SHAP values sit in the audit log.

The trust question is whether users’ trust is *calibrated* to system reliability. Jian et al. (2000)’s scale remains the most widely used instrument: twenty-five years old, which is both a strength (extensively validated) and a caveat (some items read awkwardly today). Calibrated trust matters acutely for heritage settings, where the cost of a wrong recommendation can be conservation damage. The risk of building over-trust through a polished interface that does not communicate uncertainty is well-documented; Amershi et al. (2019)’s guideline 11 (“make clear why the system did what it did”) exists precisely because users in deployed systems frequently assume reliability the system has not earned.

The gap: The HAI and interpretability literatures provide principles and techniques but not an integrated specification. Most BMS interfaces expose raw sensor data or perform automated adjustments; few combine reasoned recommendations, plain-language explanations, and auditable overrides. The components exist; the integration for a heritage building does not.

2.4 Heritage Conservation as a Design Constraint

Heritage conservation imposes constraints that most smart-building literature treats as externalities but which here must be first-order design requirements. The Burra Charter (ICOMOS Australia, 2013) and Feilden (2003) articulate principles that predate digital twins but transfer cleanly to AI. Three are load-bearing: interventions must be *minimal*, *reversible*, and *explicable*. Automated threshold-based actuation violates all three: it acts proactively, may commit the building to changes not easily reversed, and offers no intrinsic explanation. A system that does not record why it intervened cannot be audited, which for listed buildings is a statutory rather

than theoretical concern.

Mazzetto (2024) frames the heritage-AI problem as a three-way tension between conservation, energy sustainability, and occupant wellbeing, with most deployed digital twin work optimising for one at the expense of the others. Arowoia et al. (2024), reviewing the state of the art, report that across the literature they surveyed no system personalises comfort recommendations for individuals in heritage settings. This is a careful statement about *their* review, not a universal claim; such a system may exist in unpublished work their search did not capture. With that caveat, their finding is the strongest evidence the personalised-heritage gap is real.

The symmetry between active learning and the Burra Charter’s minimum-intervention principle shapes this project’s contribution. Active learning, by construction, queries the user only when the model is uncertain; otherwise the system is silent. This is the algorithmic analogue of minimum intervention: acting only when necessary, logging every action for review. Adding a human override / defer mechanism produces a system in which every intervention is uncertain-by-design, minimal-by-construction, and auditable-by-default. Articulating this as algorithmic embodiment of conservation principles, rather than policy layered on top, is the contribution claimed.

The gap: Within the approximately 25 directly relevant sources reviewed (and 40 with secondary relevance), I have not found work that builds minimum intervention into the algorithm itself. A more exhaustive systematic review at dissertation stage may surface counter-examples; the project’s claim is therefore careful: *within the literature reviewed*, this integration has not been done.

2.5 Summary and Implications

Pulling the strands together: the components of a conservation-sensitive, personalised, explainable comfort decision-support system exist in the literature but have not been assembled. Active learning addresses data efficiency (§2.2); interpretability and HAI principles address trust and override (§2.3); conservation ethics, as design constraints, address minimum intervention (§2.4); the need is established by the comfort evidence base (§2.1). Combining the four in a proof of concept, and showing the combination produces interpretable recommendations a non-technical decision-maker can use, is what this project does. *This is the gap from which the research question stated in §1.2 is derived.*

2.5.1 Limitations of This Review

The heritage-specific ML literature is thin: Mazzetto (2024) and Arowoia et al. (2024) are diagnostic rather than prescriptive. The active-learning-for-comfort literature is drawn from a small number of office-based studies. The corpus reviewed here is approximately 25 directly relevant sources; a fuller systematic review at dissertation stage may surface evidence not captured here. The project framing reflects this: it is a proof of concept, evaluated on existing data, not a deployment-ready system.

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A Project Plan

A.1 Methodological Approach

The project adopts a design-science research methodology (Hevner et al., 2004) structured around iterative development sprints. Design science recognises two complementary outputs from the same project: the **artefact** (the software prototype) and the **research evidence** obtained *during* the artefact’s development. Both are deliverables.

Artefact-side outputs. The proof-of-concept system itself, with documented architecture, source code, and a reproducibility README. Each sprint produces an evaluable intermediate artefact (literature foundation; architecture; minimum-viable prototype; final prototype with explanation layer).

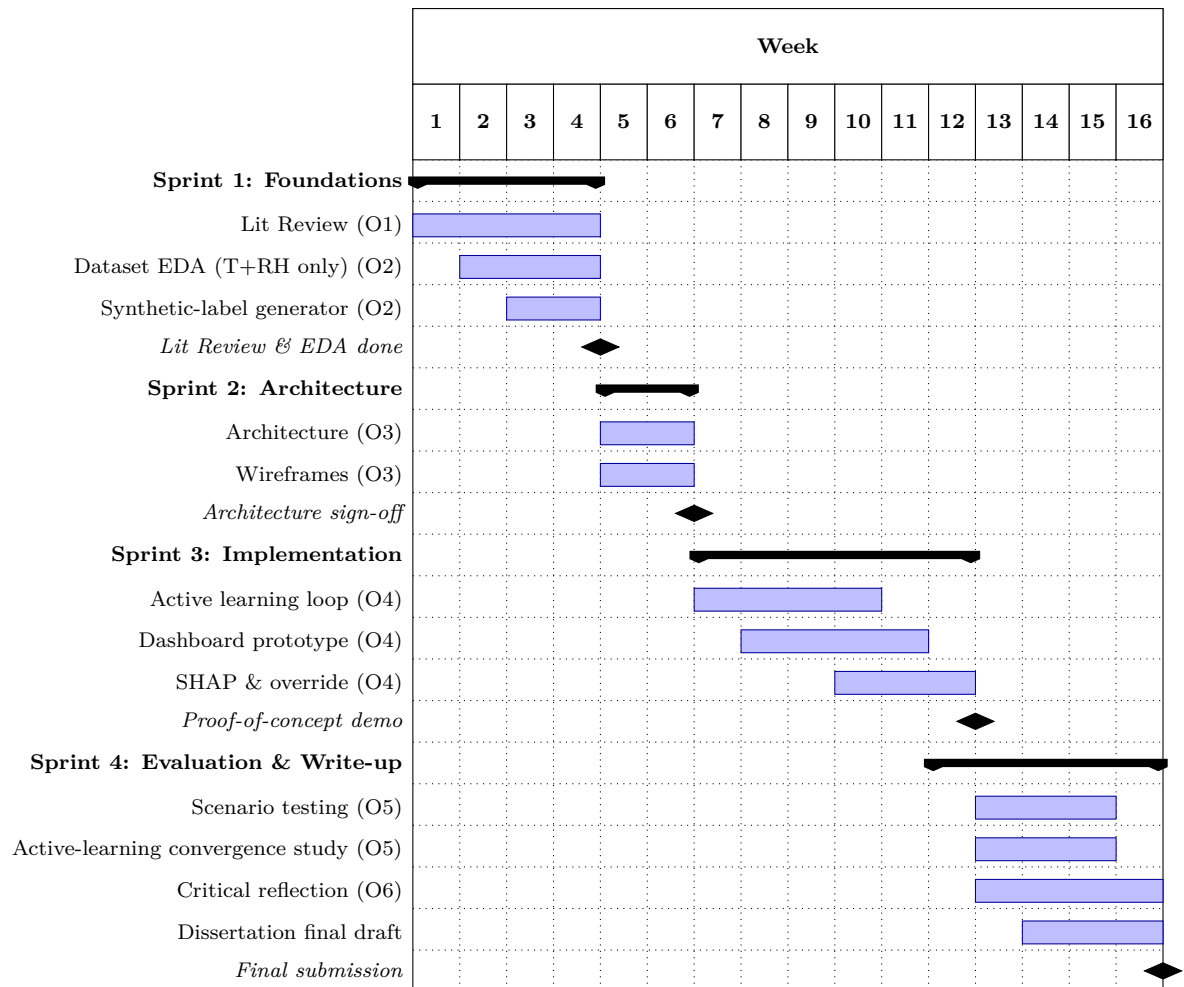
Research-side evidence collection. Three streams of evidence run alongside artefact development:

1. **Design-decision log.** Every architectural decision (uncertainty threshold value, query-strategy choice, explanation format) is recorded with its alternatives considered, the criterion used to choose, and the supervisor or expert input that informed it. This is the primary audit trail and the central evidence stream of the dissertation.
2. **Scenario-based testing.** The trained model is evaluated on a held-out set of approximately six representative LaSDPC episodes, with the recommendations it generates compared against post-hoc expert-judged “correct” actions. Specific success thresholds (e.g. $\geq 70\%$ match) are stated in the dissertation methodology chapter.
3. **Active-learning convergence study.** The model’s accuracy as a function of label budget is recorded. The thresholds set for “acceptable convergence” (e.g. within 5% of the full-labelled baseline using $\leq 40\%$ of labels) are stated explicitly.

What the project does NOT do at MSc proof-of-concept stage. Live human-subjects evaluation, including think-aloud expert walkthroughs as previously proposed, has been deferred. The reasoning is twofold. First, recruiting five conservation officers and HCI researchers is high-effort for a proof-of-concept stage where the goal is to demonstrate algorithmic feasibility rather than usability validation. Second, live evaluation with named participants would require ethical approval that the proof-of-concept stage does not justify. Both move to future-work scope, with a placeholder in the project plan for a small expert review (1–2 supervisory-team members commenting on a recorded walkthrough video) if time permits.

A.2 Sprint Plan and Gantt Chart

The project runs across one academic year (16 weeks of active development plus supervisory and write-up windows on either side) and is structured as four sprints.



A.3 Tasks and Effort Estimation

Sprint	Task (broken down into smaller units)	Start	End	Effort (h)
1	Literature review: thermal comfort strand	W1	W2	18
1	Literature review: ML for personal comfort strand	W2	W3	14
1	Literature review: HAI & heritage strands	W3	W4	18
1	Dataset EDA (T & RH only)	W2	W4	20
1	Synthetic-label generator design & test	W3	W4	10
2	System architecture & query-strategy choice	W5	W6	25
2	Interface wireframing & information hierarchy	W5	W6	20
3	Active learning loop implementation	W7	W10	50
3	Dashboard prototype (Streamlit)	W8	W11	40
3	SHAP integration & plain-language layer	W10	W12	25
3	Override / defer / confirm logic + audit log	W11	W12	12
4	Scenario-based testing on LaSDPC episodes	W13	W14	22
4	Active-learning convergence study	W13	W15	15
4	Critical reflection & dissertation write-up	W13	W16	75
Total estimated effort				364

A.4 Milestones

1. **Literature Review & Dataset EDA:** end of Week 4.
2. **System Architecture sign-off:** end of Week 6.
3. **Proof-of-concept demonstration:** end of Week 12.
4. **Final dissertation submission:** end of Week 16.

A.5 Risk Register

Risk	Prob.	Impact	Mitigation
No pre-existing comfort labels	High	Med	Synthetic feedback via PMV + Gaussian noise calibrated to Cheung et al. (2019) variance; documented as scope decision; in fact motivates the active-learning approach.
Dataset quality issues	Med	Med	Rigorous EDA in Sprint 1; mean imputation for short gaps; flag and exclude extended gaps; dataset audit report.
Active learning fails to converge on this data	Low	Med	Fallback to static Random Forest with synthetic labels (interface + evaluation contributions intact).
MVP scope creep	Med	Low	Strict feature list locked before Sprint 3; “may” requirements deferred to future work.
Illness or hardware failure	Low	Med	GitHub version control; Bath filesystem and Google Drive backups; sprint slack absorbs short absences.

A.6 Resources

Datasets (temperature and humidity only, no PII)

- **LaSDPC IoT dataset** (University of São Paulo): 12+ months of readings across building zones, available at <https://github.com/Smart-LaSDPC/iot-dataset>. This project uses the temperature and relative humidity channels only.
- **Bath heritage dataset** (Connaught Mansions): Tinytag Ultra 2 logs collected by Cai (2025). This project uses the temperature and relative humidity readings only; no occupant identifiers or detailed location information are involved.

Software

Python 3.x with scikit-learn (active learning, Random Forest); SHAP (Lundberg and Lee, 2017) for feature-importance; Streamlit for the dashboard prototype; Jupyter for EDA; GitHub for version control; Overleaf / $\text{\LaTeX}$ for write-up.

Hardware

Personal laptop sufficient for proof-of-concept scope; University HPC available as fallback.

Human resources

- **Dr Fabio Nemetz:** primary supervisor, weekly meetings.
- **Dr Ricardo Codinhoto:** co-supervisor, milestone reviews.
- **Dr Paul Shepherd:** co-supervisor, milestone reviews.
- International collaboration with USP, Unicamp and UNESP: LaSDPC dataset expertise.